

Software Requirement Specifications

SmartCalc: Algebraic Solver Edition

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1 Introduction

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements for **SmartCalc**. This application is designed to bridge the gap between standard arithmetic calculators and complex graphing tools by offering an intuitive **Linear Equation Solver**.

1.2 Scope

The software is a mobile application (Android/iOS) that allows users to:

1. Perform standard arithmetic calculations.
2. Input linear algebraic equations using a specialized keypad.
3. Receive instant solutions for the variable x .

1.3 Definitions and Acronyms

- **SRS:** Software Requirement Specification.
- **PEMDAS:** Order of operations (Parentheses, Exponents, Multiplication, Division, Addition, Subtraction).
- **Linear Equation:** An equation of the first degree (e.g., $ax + b = c$).

2 Overall Description

2.1 Product Perspective

SmartCalc operates as a standalone application. It does not require internet connectivity for its solving engine, ensuring availability in exam or field environments (subject to academic integrity policies).

2.2 User Classes

- **Students:** Primary users requiring assistance with algebra homework.
- **General Users:** Users needing quick arithmetic.

3 System Features

3.1 Standard Arithmetic Engine

Description: The system shall function as a standard calculator.

Functional Requirement 1.0: The system shall accept inputs for numbers (0-9) and operators (+, -, ×, ÷).

Functional Requirement 1.1: The system shall handle floating-point results up to 10 decimal places.

3.2 Linear Equation Solving Engine

Description: The system shall parse string inputs containing the variable 'x' and the equality operator '=' to solve for 'x'.

3.2.1 Input Constraints

- The equation must contain exactly one variable type (x).
- The equation must contain exactly one equality sign (=).
- The equation degree must not exceed 1 (Linear only, no x^2).

3.2.2 Solved Equation Examples

The system is explicitly required to solve the following mathematical patterns.

Case A: Standard Form ($ax + b = c$)

The solver must handle basic integer coefficients.

$$3x + 15 = 30 \quad (1)$$

System Output: $x = 5$

Case B: Variables on Both Sides

The solver must group like terms from both sides of the equality.

$$5x - 10 = 2x + 14 \quad (2)$$

Logic: $5x - 2x = 14 + 10 \rightarrow 3x = 24$.

System Output: $x = 8$

Case C: Parentheses & Distribution

The system must apply the distributive property before solving.

$$2(x - 3) + 4 = 10 \quad (3)$$

Logic: $2x - 6 + 4 = 10 \rightarrow 2x - 2 = 10 \rightarrow 2x = 12$.

System Output: $x = 6$

Case D: Fractional Coefficients

The system must handle rational numbers.

$$\frac{1}{2}x + \frac{1}{4} = \frac{3}{4} \quad (4)$$

System Output: $x = 1$

4 External Interface Requirements

4.1 User Interface (UI)

- **Screen 1 (Calc):** Standard numeric keypad.
- **Screen 2 (Algebra):** Extended keypad adding keys: $[x]$, $[()$, $[])]$, $[=]$.
- **Result Display:** A distinct display area showing the step, e.g., " $x = 5$ ".

5 Non-Functional Requirements

5.1 Performance

- **NFR-1:** Equation solving time shall not exceed 0.5 seconds on standard devices.
- **NFR-2:** App launch time shall be under 2 seconds.

5.2 Robustness

- **NFR-3:** If a user inputs a non-linear equation (e.g., $x^2 + 2 = 0$), the system shall return a "Feature Not Supported" error rather than crashing.
- **NFR-4:** Infinite solution cases (e.g., $2x = 2x$) shall display "Infinite Solutions".